Retrieval-Augmented Machine Translation via k-Nearest Neighbor Machine Translation (kNN-MT)

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https://github.com/zedaeda/rag_machine_translation

Abstract. Neural machine translation (NMT) achieves strong performance on general-domain text but struggles in specialized or domain-specific settings. This paper presents an implementation of k-Nearest Neighbor Machine Translation (kNN-MT), a retrieval-augmented approach that augments a pre-trained NMT model with a non-parametric datastore at inference time without any additional training. We train a Transformer NMT baseline on the IWSLT 2017 English-German dataset and construct Faiss-based datastores from two corpora: the in-domain IWSLT training set and the cross-domain Europarl parliamentary proceedings corpus. At decoding time, the kNN distribution is interpolated with the NMT distribution using a mixing coefficient λ. We conduct ablation studies on λ, the number of neighbors K, and the temperature T. Our results demonstrate that the cross-domain Europarl datastore consistently outperforms the in-domain IWSLT datastore across all six evaluation metrics. Furthermore, integrating kNN-MT with beam search achieves 29.24 BLEU, approaching the 32.10 BLEU beam search baseline, confirming that retrieval-augmented beam search is feasible within the fairseq framework.

Keywords: Machine Translation · Retrieval-Augmented Generation · kNN-MT · Faiss · IWSLT · Europarl

1 Introduction

Neural machine translation has achieved remarkable results on general-domain benchmarks, yet domain adaptation remains a persistent challenge. Fine-tuning large NMT models on domain-specific data is expensive and requires substantial in-domain parallel corpora. kNN-MT [1] addresses this by augmenting a frozen NMT model with a retrieval mechanism: at each decoding step, the decoder hidden state is used to query a datastore of (context representation, target token) pairs, producing a non-parametric distribution that is interpolated with the NMT output distribution.

This work implements kNN-MT on the IWSLT 2017 English-German spoken language translation benchmark. We train a Transformer NMT baseline using the fairseq toolkit [3], construct Faiss [4] datastores from both the IWSLT training set and the Europarl corpus, and evaluate the full system using BLEU [6], chrF, TER, METEOR, ROUGE-L, and BERTScore. We conduct ablation studies on key hyperparameters, provide qualitative error analysis, and implement beam search integration with kNN interpolation via output logit injection. A key finding is that the cross-domain Europarl datastore produces consistent improvements over the in-domain IWSLT datastore, validating the core hypothesis of kNN-MT.

2 Related Work

Transformer NMT. The Transformer architecture [2] is the foundation of modern NMT systems. Its encoder-decoder structure with multi-head self-attention produces strong translation quality on large parallel corpora.

kNN-MT. Khandelwal et al. [1] proposed augmenting NMT with a nearest neighbor retrieval mechanism. At inference time, decoder hidden states are used to query a Faiss index of training-set representations. The retrieved neighbors vote on the next token, and their distribution is mixed with the NMT distribution via a fixed interpolation weight λ. The method demonstrates gains of up to 16 BLEU points in cross-domain settings where the base model is trained on a different domain than the datastore.

Efficient retrieval. Johnson et al. [4] introduced Faiss, a library for efficient similarity search over dense vectors. Meng et al. [5] proposed pruning strategies to accelerate kNN-MT inference without significant quality loss.

Evaluation. BLEU [6] remains the standard MT metric; Post [7] motivates the use of sacrebleu for reproducible reporting. BERTScore and METEOR provide complementary semantic evaluation signals. chrF measures character-level F-score and is particularly suited for morphologically rich languages such as German. TER measures the number of edits required to transform the hypothesis into the reference.

3 Methodology

3.1 Baseline: Transformer NMT

We train a standard Transformer sequence-to-sequence model on IWSLT 2017 En-De using fairseq [3]. The architecture follows the `transformer_iwslt_de_en` configuration: 6 encoder and decoder layers, 4 attention heads, embedding dimension 512. A shared SentencePiece BPE vocabulary of 10,000 tokens is trained on the combined training corpus. The model is trained for 32 epochs with early stopping (patience=5), Adam optimizer, inverse square root learning rate schedule, and label smoothing of 0.1.

3.2 Datastore Construction

Following Khandelwal et al. [1], we construct token-level datastores by performing teacher-forced forward passes through the trained NMT decoder and extracting hidden states from the final decoder layer after its layer normalization (`layers[-1].final_layer_norm`). These hidden states form the *keys* of the datastore, while the corresponding target tokens form the *values*. Keys are indexed using a Faiss `IndexFlatL2` exact search index.

We construct two datastores: (1) an **in-domain IWSLT datastore** containing 206,112 (key, value) pairs from the IWSLT training set, and (2) a **cross-domain Europarl datastore** containing 206,112 pairs sampled from the Europarl En-De parliamentary proceedings corpus [8]. Both datastores use the same BPE vocabulary and model checkpoint, ensuring a controlled comparison.

3.3 kNN-MT Inference

At each decoding step t, the decoder produces a hidden state $h_t \in \mathbb{R}^{512}$ and a softmax distribution p_{NMT} over the vocabulary. We query the Faiss index to retrieve the K nearest neighbors of h_t by L2 distance, obtaining distances $d_1, \ldots, d_K$ and corresponding target tokens $v_1, \ldots, v_K$. The kNN distribution is computed as:

$$p_{\mathrm{kNN}}(y) \propto \sum_{i: v_i = y} \exp\left(-d_i / T\right) \tag{1}$$

The final distribution is interpolated as:

$$p_{\mathrm{final}} = \lambda \cdot p_{\mathrm{kNN}} + (1 - \lambda) \cdot p_{\mathrm{NMT}} \tag{2}$$

where λ is the interpolation weight and T is the temperature parameter controlling the sharpness of the kNN distribution.

3.4 Beam Search Integration

To integrate kNN-MT with beam search, we implement output logit injection via fairseq's forward hook mechanism. At each decoding step, we attach a hook to the decoder's output projection layer (`output_projection`) to intercept the logits before beam selection. A second hook on `layers[-1].final_layer_norm` captures the hidden state for each active beam hypothesis. The kNN log-probability distribution is computed for each beam independently and combined with the NMT log-probabilities:

$$\log p_{\mathrm{final}} = (1 - \lambda) \cdot \log p_{\mathrm{NMT}} + \lambda \cdot \log p_{\mathrm{kNN}} \tag{3}$$

This approach requires no modifications to fairseq's internal beam search loop, as the injection occurs transparently at the logit level.

3.5 Implementation Challenges

Unavailability of the WMT14 pretrained checkpoint. The original paper applies kNN-MT on top of a WMT14-pretrained fairseq model. The pretrained checkpoint hosted at `dl.fbaipublicfiles.com` returned HTTP 403 Forbidden, as Meta/Facebook's model server has been taken offline. Alternative sources were also unavailable or format-incompatible. As a result, kNN-MT was applied on top of the IWSLT-trained Baseline 1 model. To partially compensate, we constructed a cross-domain Europarl datastore that introduces vocabulary and style not seen during IWSLT training.

Repetition loops in kNN-MT output. Initial runs produced severe repetition loops. The root cause was double tokenization: fairseq's `model.encode()` function was re-tokenizing already-tokenized input, producing corrupted hidden states. The fix was a custom `encode_tok_line()` function that maps BPE tokens directly to fairseq dictionary IDs, bypassing the tokenizer entirely.

Google Colab resource constraints. Training and inference were conducted on Google Colab T4 GPU (15.6 GB VRAM). Session disconnections and daily GPU quotas required careful checkpoint management via Google Drive integration.

4 Experimental Setup

Code. Full implementation available at: `https://github.com/zedaeda/rag_machine_translation`

Dataset. IWSLT 2017 English-German: 206,112 training pairs, 888 validation pairs, 8,079 test pairs (TED talk transcripts). Europarl En-De: 206,112 pairs sampled from 1.9 million parliamentary proceedings sentence pairs.

Tokenization. SentencePiece BPE, vocabulary size 10,000, trained on IWSLT combined En-De training data. The same vocabulary is applied to Europarl to ensure compatibility with the trained model.

Evaluation metrics. BLEU (SacreBLEU [7], mixed case, tokenizer 13a), chrF, TER, METEOR, ROUGE-L, and BERTScore F1 (multilingual BERT).

Hardware. Google Colab T4 GPU (15.6 GB VRAM).

kNN-MT default settings. $K = 8$, $\lambda = 0.05$, $T = 10$.

5 Results

Table 1 presents the main results across all six evaluation metrics. Baseline 1 with beam search (beam=5) achieves 32.10 BLEU and 0.8721 BERTScore, representing the strongest reference. The greedy kNN-MT system with the IWSLT datastore achieves 13.04 BLEU, slightly below the greedy NMT baseline (13.74). The Europarl cross-domain datastore improves over the IWSLT datastore on all six metrics: BLEU increases from 13.04 to 14.22 (+1.18), TER decreases from 118.13 to 111.45 (-6.68), and BERTScore improves from 0.7582 to 0.7671. Most notably, integrating kNN-MT with beam search using the Europarl datastore achieves 29.24 BLEU, approaching the 32.10 BLEU beam search baseline with a gap of only 2.86 BLEU points.

Table 1. Main results on IWSLT 2017 En-De test set (500 sentences). TER: lower is better; all others: higher is better.

System	BLEU	chrF	TER↓	METEOR	ROUGE-L	BERTScore
Baseline 1 (beam=5)	32.10	57.71	54.98	0.5216	0.5937	0.8721
Baseline 1 (greedy)	13.74	40.92	115.76	0.3437	0.3900	0.7654
kNN-MT + IWSLT (greedy)	13.04	39.48	118.13	0.3310	0.3785	0.7582
kNN-MT + Europarl (greedy)	14.22	41.36	111.45	0.3461	0.3929	0.7671
kNN-MT + Europarl (beam=5)	**29.24**	**57.31**	**61.74**	**0.5128**	**0.5636**	**0.8556**

5.1 Ablation Study

Interpolation weight λ. Table 2 shows the effect of λ (200 test sentences, K=8, T=10, IWSLT datastore). The optimal value is λ=0.05; performance degrades sharply for $\lambda \geq 0.2$.

Table 2. Ablation: BLEU vs. λ (K=8, T=10, 200 sentences).

λ	BLEU ↑
0.00	13.21
0.05	**13.63**
0.10	12.90
0.20	5.66
0.30	1.94

Number of neighbors K. Table 3 shows the effect of K (λ=0.05, T=10). Performance is stable for $K \geq 8$.

Table 3. Ablation: BLEU vs. K (λ=0.05, T=10, 200 sentences).

K	BLEU ↑
4	13.61
8	**13.63**
16	13.63
32	13.63

Temperature T. Table 4 shows the effect of T (λ=0.05, K=8). Lower temperature produces sharper kNN distributions and consistently better results.

Table 4. Ablation: BLEU vs. T (λ=0.05, K=8, 200 sentences).

T	BLEU ↑
10	**13.63**
50	13.57
100	13.54
200	13.41

6 Error Analysis

6.1 IWSLT Datastore vs. Greedy Baseline

We compare kNN-MT (IWSLT datastore) and greedy baseline outputs on 500 test sentences using sentence-level BLEU. kNN-MT improves over the baseline on 40 sentences (8%), performs worse on 429 sentences (86%), and ties on 31 sentences (6%).

Cases where kNN-MT helps. kNN-MT improves short, common phrases where the datastore provides strong retrieval signal. For example:

- SRC: *Who does the very best?*
  BL1: *Wer macht das Beste?* (BLEU: 12.7)
  kNN: *Wer ist am besten?* (BLEU: 32.5)

Cases where kNN-MT hurts. For longer, complex sentences, the kNN distribution occasionally introduces repetition or malformed output, degrading fluency under greedy decoding.

Discussion. The predominance of losses reflects the in-domain limitation: the model already knows these patterns, so retrieval adds little new information.

6.2 Europarl Datastore vs. Greedy Baseline

We compare kNN-MT (Europarl datastore) and greedy baseline outputs on 500 test sentences. The Europarl kNN-MT system wins on 15 sentences (3%), loses on 10 sentences (2%), and ties on 475 sentences (95%).

Cases where Europarl kNN-MT wins. The cross-domain datastore helps in cases where the baseline produces severe repetition loops. For example:

- SRC: *And any team who manages and pays close attention to work will significantly improve...*
  BL1: *the the the the the...* (BLEU: 0.0)
  EP: *Und jedes Team, das genau funktioniert, wird die Leistung des Teams steigern...* (BLEU: 26.9)

Cases where Europarl kNN-MT loses. Minor losses occur on very short sentences where small surface differences determine BLEU. For example:

– SRC: *And it's pretty amazing.*
  BL1: *the the the ... icicicicici* (BLEU: 10.3)
  EP: *the the the the ist erstaunlich.* (BLEU: 8.3)

Discussion. The 95% tie rate confirms that the Europarl datastore brings kNN-MT to parity with the greedy baseline, a significant improvement over the IWSLT setup where losses dominated at 86%.

7 Discussion

In the greedy setup, the Europarl kNN-MT system (14.22 BLEU) outperforms both the IWSLT kNN-MT system (13.04) and the greedy baseline (13.74), demonstrating that cross-domain retrieval provides genuine benefit even under greedy decoding. Integrating kNN-MT with beam search using the Europarl datastore achieves 29.24 BLEU, approaching the 32.10 BLEU beam search baseline. The remaining gap of 2.86 BLEU points reflects the domain mismatch between the Europarl datastore and the IWSLT test set.

The WMT14 pretrained checkpoint was unavailable (HTTP 403). To partially replicate the cross-domain setup, we used Europarl as an alternative. While the improvement is smaller than the 16 BLEU points reported in the original paper — which used a much larger WMT-pretrained model — the direction is consistent: cross-domain datastores outperform in-domain ones. The error analysis confirms this: IWSLT datastore loses on 86% of sentences, while Europarl drops to only 2% losses with 95% parity.

Future work includes: (1) obtaining a WMT-scale pretrained model for closer replication of the original paper; (2) approximate Faiss indices (IVF-PQ) for faster inference at scale.

8 Conclusion

We implemented kNN-MT on IWSLT 2017 En-De with two datastores and beam search integration. The cross-domain Europarl datastore consistently outperforms the in-domain IWSLT datastore across all six metrics (BLEU: 14.22 vs. 13.04, BERTScore: 0.7671 vs. 0.7582), validating the core retrieval hypothesis of kNN-MT. Beam search integration with the Europarl datastore achieves 29.24 BLEU, approaching the 32.10 BLEU beam search baseline and demonstrating that retrieval-augmented beam search is feasible within the fairseq framework via output logit injection. Error analysis reveals a striking contrast: IWSLT datastore loses to the greedy baseline on 86% of sentences, while Europarl achieves 95% parity. Ablation studies show sensitivity to λ but robustness to K, with lower temperature consistently improving performance.

Ethical Considerations

The kNN datastore memorizes training-set patterns explicitly, which may propagate biases present in the source corpus. The IWSLT corpus reflects the demographics of TED speakers; the Europarl corpus reflects European parliamentary discourse. Both may contain gender-associated pronoun choices or cultural framing. Hallucination risks exist when retrieved neighbors introduce contextually plausible but semantically incorrect tokens. This system is not suitable for certified translation workflows without human post-editing. Automatic metrics do not capture all dimensions of translation quality such as pragmatics or cultural appropriateness.

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